

main.c

```
#include <Arduino.h>

#define RXD2 16
#define TXD2 17
#define DE_RE 4

const uint8_t comm[3][6] = {
    {0x01, 0x04, 0x00, 0x01, 0x00, 0x01},
    {0x01, 0x04, 0x00, 0x02, 0x00, 0x01},
    {0x01, 0x04, 0x00, 0x01, 0x00, 0x02}
};

const uint8_t resp_size[3] = {7, 7, 9};

uint8_t comando[8];
uint8_t resposta[9];

uint16_t calculateCRC(uint8_t *frame, int tamanho);
unsigned char getResp();

//=====//

void setup() {
    Serial.begin(9600);
    Serial2.begin(9600, SERIAL_8N1, RXD2, TXD2);

    pinMode(DE_RE, OUTPUT);
    digitalWrite(DE_RE, LOW);
    delay(1000);
}

//=====//

void loop() {
    unsigned char resp = 0;

    while (resp < '0' || resp > '2'){
        resp = getResp();
    }

    resp -= '0';

    for (int i = 0; i < 6; i++){
        comando[i] = comm[resp][i];
    }

    uint16_t crc = calculateCRC(comando, 6);
    comando[6] = crc & 0xFF;
    comando[7] = crc >> 8;

    while (Serial2.available()){
        Serial2.read();
    }

    digitalWrite(DE_RE, HIGH);
    delay(2);

    Serial2.write(comando, sizeof(comando));
    Serial2.flush();

    digitalWrite(DE_RE, LOW);
    delay(200);

    int j = 0;
    while (Serial2.available() && j < resp_size[resp]) {
        resposta[j] = Serial2.read();
        j++;
    }

    if (j != resp_size[resp]) return;

    crc = calculateCRC(resposta, resp_size[resp] - 2);

    if (resposta[resp_size[resp] - 2] != (crc & 0xFF) ||
        resposta[resp_size[resp] - 1] != (crc >> 8)) return;

    if (resp == 0 || resp == 1) {
        uint16_t valor = (resposta[3] << 8) | resposta[4];
        Serial.println(valor);
    }

    else if (resp == 2) {
        uint16_t temperatura = (resposta[3] << 8) | resposta[4];
        uint16_t umidade = (resposta[5] << 8) | resposta[6];

        Serial.print(temperatura);
        Serial.print(":");
        Serial.println(umidade);
    }
}

//=====//

unsigned char getResp() {
    unsigned char dado = 0;

    if (Serial.available() > 0)
        dado = Serial.read();
    delay(100);
}
```

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    return dado;
}

//=====//

uint16_t calculateCRC(uint8_t *frame, int tamanho) {
    uint16_t crc = 0xFFFF;

    for (int n = 0; n < tamanho; n++) {
        crc ^= frame[n];

        for (int m = 0; m < 8; m++) {
            if (crc & 0x0001) {
                crc >>= 1;
                crc ^= 0xA001;
            }

            else {
                crc >>= 1;
            }
        }
    }

    return crc;
}

```